Year 10 Foundation

REVISION PACK

Progress Check 2

ANSWERS

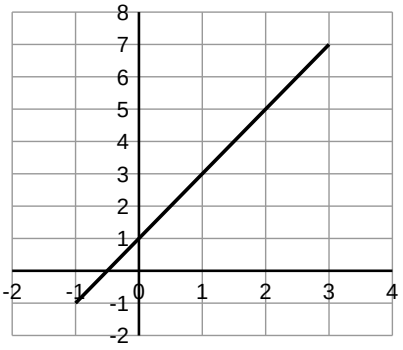
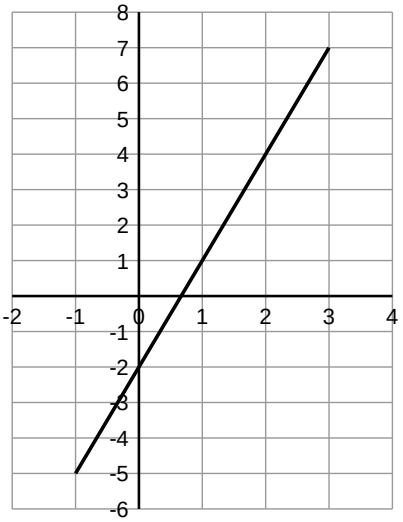
Units in this pack

- Graphs 1
- Number 2
- Shape and Space 1

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Graphs 1 · Gradients and straight-line graphs (Lessons 3.1, 3.2, 3.3)

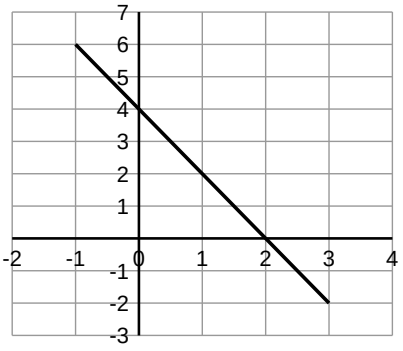
Q	Working	Answer
gradient from a graph		
1.1	$(0, -2)$ and $(2, 4)$ up 6, across 2 $\frac{6}{2} = 3$	3
1.2	$(-1, 5)$ and $(2, -4)$ up -9, across 3 $\frac{-9}{3} = -3$	-3
1.3	$(0, 1)$ and $(4, 3)$ up 2, across 4 $\frac{2}{4} = \frac{1}{2}$	$\frac{1}{2}$
1.4	$(-2, -4)$ and $(1, 5)$ up 9, across 3 $\frac{9}{3} = 3$	3
gradient from two points		
1.5	$\frac{18 - 3}{4 - 1} = \frac{15}{3}$ $= 5$	5
1.6	$\frac{1 - 10}{5 - 2} = \frac{-9}{3}$ $= -3$	-3
1.7	$\frac{6 - 2}{8 - 0} = \frac{4}{8}$ $= \frac{1}{2}$	$\frac{1}{2}$
1.8	$\frac{16 - 4}{3 - (-1)} = \frac{12}{4}$ $= 3$	3

1.9	$\frac{-1-7}{6-2} = \frac{-8}{4}$ $= -2$	-2
1.10	$\frac{10-(-2)}{5-1} = \frac{12}{4}$ $= 3$	3
drawing a straight-line graph		
1.11	$x = -1: \ y = 2 \times (-1) + 1 = -1$ <table><tr><td>x</td><td>−1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>y</td><td>−1</td><td>1</td><td>3</td><td>5</td><td>7</td></tr></table> 	**(a) y = −1, 1, 3, 5, 7** **(b) a straight line through the points**
1.12	$x = -1: \ y = 3 \times (-1) - 2 = -5$ <table><tr><td>x</td><td>−1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>y</td><td>−5</td><td>−2</td><td>1</td><td>4</td><td>7</td></tr></table> 	**(a) y = −5, −2, 1, 4, 7** **(b) a straight line through the points**

1.13

$x = -1 : \ y = -2 \times (-1) + 4 = 6$

x	−1	0	1	2	3
y	6	4	2	0	−2

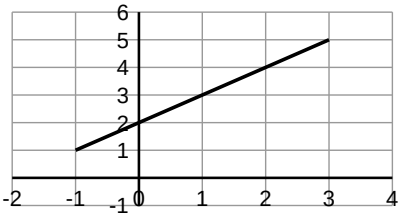


(a) y = 6, 4, 2, 0, −2
(b) a straight line through the points

1.14

$x = -1 : \ y = 1 \times (-1) + 2 = 1$

x	−1	0	1	2	3
y	1	2	3	4	5



(a) y = 1, 2, 3, 4, 5
(b) a straight line through the points

gradient and intercept

1.15

(a) $m = 4$

(b) $c = -3$

(a) 4
(b) (0, −3)

1.16

(a) $m = -5$

(b) $c = 8$

(a) −5
(b) (0, 8)

1.17

(a) $m = -1$

(b) $c = 6$

(a) −1
(b) (0, 6)

1.18

(a) $m = 3$

(b) $c = 2$

(a) 3
(b) (0, 2)

1.19	**(a)** $m = -4$ **(b)** $c = 9$	**(a) −4** **(b) (0, 9)**
1.20	**(a)** $m = 6$ **(b)** $c = -1$	**(a) 6** **(b) (0, −1)**

Number 2 · Percentages (Lesson 4.1)

Q	Working	Answer
percentage decrease		
2.1	12% of $250 = 30$ $250 - 30 = 220$	**KD 220**
2.2	25% of $640 = 160$ $640 - 160 = 480$	**KD 480**
2.3	40% of $85 = 34$ $85 - 34 = 51$	**KD 51**
2.4	15% of $320 = 48$ $320 - 48 = 272$	**KD 272**
2.5	30% of $150 = 45$ $150 - 45 = 105$	**KD 105**
2.6	25% of $96 = 24$ $96 - 24 = 72$	**KD 72**
percentage loss		
2.7	$5000 - 4250 = 750$ $\frac{750}{5000} \times 100 = 15\%$	**15%**
2.8	$800 - 680 = 120$ $\frac{120}{800} \times 100 = 15\%$	**15%**
2.9	$120 - 102 = 18$ $\frac{18}{120} \times 100 = 15\%$	**15%**

2.10	$2400 - 1800 = 600$ $\frac{600}{2400} \times 100 = 25\%$	**25%**
2.11	$650 - 520 = 130$ $\frac{130}{650} \times 100 = 20\%$	**20%**
2.12	$9000 - 7650 = 1350$ $\frac{1350}{9000} \times 100 = 15\%$	**15%**

Shape and Space 1 · Angles (Lessons 5.1, 5.2)

Q	Working	Answer
the exterior angle of a triangle		
3.1	$x = 47 + 68 = 115$	$x = 115°$**. The exterior angle of a triangle equals the sum of the two interior opposite angles.**
3.2	$x = 39 + 82 = 121$	$x = 121°$**. The exterior angle of a triangle equals the sum of the two interior opposite angles.**
3.3	$x = 55 + 64 = 119$	$x = 119°$**. The exterior angle of a triangle equals the sum of the two interior opposite angles.**
3.4	$x = 61 + 45 = 106$	$x = 106°$**. The exterior angle of a triangle equals the sum of the two interior opposite angles.**
3.5	$x = 33 + 90 = 123$	$x = 123°$**. The exterior angle of a triangle equals the sum of the two interior opposite angles.**
3.6	$x = 58 + 71 = 129$	$x = 129°$**. The exterior angle of a triangle equals the sum of the two interior opposite angles.**
angles in a quadrilateral		
3.7	$x = 360 - 96 - 81 - 92$	$x = 91°$

3.8	$x = 360 - 115 - 70 - 88$	$x = 87°$
3.9	$x = 360 - 78 - 108 - 95$	$x = 79°$
3.10	$x = 360 - 100 - 85 - 90$	$x = 85°$
3.11	$x = 360 - 120 - 65 - 80$	$x = 95°$
3.12	$x = 360 - 88 - 94 - 110$	$x = 68°$

Shape and Space 1 · Pythagoras' theorem, area and circles (Lessons 5.3, 5.4, 5.5)

Q	Working	Answer
finding the hypotenuse		
4.1	$x^2 = 12^2 + 16^2 = 400$ $x = \sqrt{400} = 20$	x **= 20 cm**
4.2	$x^2 = 5^2 + 12^2 = 169$ $x = \sqrt{169} = 13$	x **= 13 cm**
4.3	$x^2 = 7^2 + 10^2 = 149$ $x = \sqrt{149} = 12.2066\ldots$	x **= 12.2 cm**
4.4	$x^2 = 8^2 + 15^2 = 289$ $x = \sqrt{289} = 17$	x **= 17 cm**
4.5	$x^2 = 9^2 + 11^2 = 202$ $x = \sqrt{202} = 14.2127\ldots$	x **= 14.2 cm**
4.6	$x^2 = 20^2 + 21^2 = 841$ $x = \sqrt{841} = 29$	x **= 29 cm**
compound shapes		
4.7	$\text{rectangle} = 14 \times 6 = 84$ $\text{triangle} = \frac{1}{2} \times 14 \times 4 = 28$ $\text{total} = 112$	**112 cm²**
4.8	$\text{rectangle} = 10 \times 7 = 70$ $\text{triangle} = \frac{1}{2} \times 10 \times 5 = 25$ $\text{total} = 95$	**95 cm²**

4.9	$\text{rectangle} = 16 \times 5 = 80$ $\text{triangle} = \frac{1}{2} \times 16 \times 3 = 24$ $\text{total} = 104$	**104 cm²**
4.10	$\text{rectangle} = 8 \times 6 = 48$ $\text{triangle} = \frac{1}{2} \times 8 \times 4 = 16$ $\text{total} = 64$	**64 cm²**
4.11	$\text{rectangle} = 18 \times 4 = 72$ $\text{triangle} = \frac{1}{2} \times 18 \times 6 = 54$ $\text{total} = 126$	**126 cm²**
4.12	$\text{rectangle} = 11 \times 9 = 99$ $\text{triangle} = \frac{1}{2} \times 11 \times 2 = 11$ $\text{total} = 110$	**110 cm²**
trapezium problems		
4.13	$\text{area} = \frac{1}{2}(9 + 15) \times 6 = 72$ $72 \div 20 = 3.6$ round up	**4 boxes**
4.14	$\text{area} = \frac{1}{2}(12 + 18) \times 8 = 120$ $120 \div 25 = 4.8$ round up	**5 boxes**
4.15	$\text{area} = \frac{1}{2}(7 + 13) \times 9 = 90$ $90 \div 25 = 3.6$ round up	**4 boxes**
4.16	$\text{area} = \frac{1}{2}(6 + 14) \times 5 = 50$ $50 \div 15 = 3.333$ round up	**4 boxes**
4.17	$\text{area} = \frac{1}{2}(11 + 17) \times 10 = 140$ $140 \div 30 = 4.667$ round up	**5 boxes**
4.18	$\text{area} = \frac{1}{2}(5 + 9) \times 8 = 56$ $56 \div 10 = 5.6$ round up	**6 boxes**

area of a circle		
4.19	$r = 12 \div 2 = 6$ $\pi \times 6^2 = 113.097\ldots$	**113 cm²**
4.20	$r = 18 \div 2 = 9$ $\pi \times 9^2 = 254.469\ldots$	**254 cm²**
4.21	$r = 9 \div 2 = 4.5$ $\pi \times 4.5^2 = 63.6173\ldots$	**63.6 cm²**
4.22	$r = 16 \div 2 = 8$ $\pi \times 8^2 = 201.062\ldots$	**201 cm²**
4.23	$r = 7 \div 2 = 3.5$ $\pi \times 3.5^2 = 38.4845\ldots$	**38.5 cm²**
4.24	$r = 22 \div 2 = 11$ $\pi \times 11^2 = 380.133\ldots$	**380 cm²**